Sober Baseline Protocol

The control condition for the 650 nm laser diffraction observation protocol | DMT Code research registry

PURPOSE

A registry of observations is only interpretable against a control. This protocol produces that control: the same apparatus, the same alignment, the same field sheet, recorded by sober observers. Baseline records establish what the projected diffraction pattern and its laser speckle actually look like to an unaided, unaltered visual system. Every later comparison in the registry depends on these records existing in volume. Anyone with the rig can contribute a baseline record tonight. No other condition is required.

The observation was first described by Danny Goler in August 2020 and written up as a pilot study in IPI Letters (DOI 10.59973/ipil.158); he has no part in Meridian Optics Lab, the store or the site, is not a founder, holds no editorial role and endorses nothing sold or published here.

BASELINE CONDITION, DEFINITION

For a session to qualify as a baseline record, every observer attests to all of the following for the 24 hours before the session: no psychoactive substances of any kind, including alcohol and cannabis; no new prescription medication started within the prior week; normal sleep the previous night (self-assessed); no current migraine, aura, or acute visual disturbance. Caffeine and nicotine are permitted but must be recorded. Sessions that do not meet the definition are still valuable: record them on the standard field sheet instead, and do not label them baseline.

STANDARD APPARATUS, IDENTICAL TO THE FIELD SHEET

1. 650 nm red laser pointer (Class IIIa / 3R, at or below 5 mW, the Class 3R ceiling; the kit pointers are vendor rated 5 mW, FDA Class IIIa) on an adjustable stand. Hands-free mounting required.
2. Diffraction grating in the beam path, close to the aperture, fanning the beam into the observation field. The $199 Protocol kit ships a glass rod in place of a grating, a cylindrical optic that refracts the dot into a single line and produces no diffraction orders; in that case record the Grating in beam path field as none, glass rod, and leave the diffraction line fields as none.
3. Projection surface: flat, light-colored, matte wall or screen, 3 to 5 meters from the laser (site protocol, wider than the paper; the pilot study places the surface 4 to 6 ft, 1.2 to 1.8 m, from the laser).
4. Room darkened. The pattern should be the brightest object in the visual field.

BASELINE PROCEDURE

1. Align the rig per the field sheet standard alignment procedure. Pattern spread 0.5 to 1.5 meters wide (site protocol, wider than the paper).
2. Each observer completes the sobriety attestation on page 2 before the lights go down.
3. Observe the projected pattern continuously for a timed period of at least 10 minutes, seated 1 to 3 meters from the surface (site protocol, wider than the paper). Record nothing during the first two minutes; the two minute minimum before recording is site protocol. Do not confer during the observation period.
4. Record everything observed on pages 2 and 3, using description only. Characterize the speckle field honestly: graininess, shimmer, apparent motion at the edge of fixation, afterimage behavior on blinking.
5. If any structured, repeating, or glyph-like element is observed while sober, that is a significant record. Sketch it on page 3 exactly as seen and note the time. Do not talk yourself out of it and do not embellish it.
6. Submit the completed record at dmtcode.com/registry and mark the session type as baseline.

WHY SOBER RECORDS CARRY THE MOST WEIGHT

Laser speckle is a real optical phenomenon with real perceptual texture. Untrained reports often mistake its shimmer for structure. A large body of sober records defines the boundary between what the optics put on the wall and what any altered report claims beyond it. Contributing a baseline is the single highest-leverage act available to the registry. Adults 18+ only. Read the safety card on page 4 before setup.

Baseline Session Record

Complete before and during the observation period. One sheet per observer per session.

SOBRIETY ATTESTATION, REQUIRED FOR BASELINE STATUS

- ☐ No psychoactive substances, 24 h
- ☐ No alcohol, 24 h
- ☐ No cannabis, 24 h
- ☐ No new medication, 7 days
- ☐ Normal sleep last night
- ☐ No migraine or visual disturbance

Caffeine in last 6 h (amount, time) _____

Nicotine in last 6 h (amount, time) _____

Observer signature or initials _____

Date _____

SESSION

Date _____

Start time / end time _____

Location type (home, studio, outdoor, other) _____

Observer ID or initials (optional) _____

Observers present (count) _____

Session number in your series _____

Timed observation period (minutes) _____

Session type: BASELINE _____

APPARATUS CONFIGURATION

Laser model / source _____

Wavelength (nm) _____

Grating in beam path (type / lines per mm; or glass rod) _____

Laser-to-surface distance (m) _____

Diffraction lines (count, spacing, brightness; or none) _____

Optic-to-laser distance (cm) _____

Projected pattern width on surface (m) _____

Observer-to-surface distance (m) _____

Projection surface (wall paint, screen, other) _____

Ambient light (fully dark, dim, other) _____

Corrective lenses worn (none, glasses, contacts) _____

Mount stable and hands-free (Y/N) _____

SPECKLE FIELD CHARACTER, CHECK ALL OBSERVED

- ☐ Fine granular texture
- ☐ Shimmer or boiling motion
- ☐ Motion tied to own eye movement
- ☐ Stable when fixation held
- ☐ Afterimages on blinking
- ☐ Color fringing beyond red
- ☐ Structured or repeating elements
- ☐ Glyph-like elements
- ☐ No structure beyond speckle

OBSERVATION NOTES, DESCRIPTION ONLY

Baseline Sketch Record

Sketch anything structured immediately. If nothing structured appears, sketch the speckle texture itself.

Element 1

During observation [ ] From recall [ ]

Element 2

During observation [ ] From recall [ ]

Element 3

During observation [ ] From recall [ ]

Element 4

During observation [ ] From recall [ ]

Element 5

During observation [ ] From recall [ ]

Element 6

During observation [ ] From recall [ ]

FULL-FIELD SKETCH, THE PROJECTED PATTERN AS A WHOLE

Confidence that sketches reflect what was observed (low / moderate / high) and why _____

Laser Safety Card

Keep with the apparatus. Review before every session.

THE INSTRUMENT

This protocol uses a Class IIIa / 3R visible laser, at or below 5 mW output at 650 nm, the Class 3R ceiling, compliant with US FDA CDRH requirements for consumer laser products. The kit pointers are vendor rated 5 mW, FDA Class IIIa (Class 3R). The pilot study's own instrument was a 650 nm Class 2 laser at 1 mW, quoted here for comparison only; the kits are not that setup. At this class, the blink reflex normally protects against momentary accidental exposure, but deliberate or prolonged direct viewing can cause permanent eye injury.

ABSOLUTE RULES

- Never look into the laser aperture, powered or not.
- Never point the beam at eyes, faces, people, or animals.
- Never view the beam or its reflections through magnifying optics, binoculars, cameras with optical viewfinders, or any lens system. Optics concentrate the beam and defeat the eye's natural protection.
- Never direct the beam at vehicles or aircraft. Doing so is a federal crime in the United States.
- Never leave the laser powered and unattended.
- Remove reflective objects (mirrors, glass, polished metal, dark screens) from the beam path before powering on.
- Adults 18+ operate the laser. Keep it out of the reach of children.

SESSION HYGIENE

Observers view the projected pattern on the surface, never the aperture and never specular reflections. Power the laser off between observation periods. Confirm stable mounting before each power-on so the beam cannot swing across the room if the stand is bumped. If any observer reports afterimages persisting more than a few minutes, end the session; persistent visual changes after any suspected direct exposure warrant a prompt eye examination.

CARE

Store the laser with batteries removed. Keep the grating clean and dry; handle it by the edges only. Inspect the stand and mount for looseness before each session. Replace any component that no longer holds the beam steady.